Comptia A+

Assignment A+ - Understanding and maintenance of Networks

1. Log in to your home Wi-Fi router's admin panel using its default IP address (usually 92.168.0.1 or 192.168.1.1) and take a screenshot of the login page If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'

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Objective: Access the home Wi-Fi router's administrative web interface via its local IP address

❖ **Locating the Default Gateway:**

- Opened the Command Prompt (cmd) on Windows.
- Executed the ipconfig command to display all network interfaces.
- Identified the IPv4 address under **Default Gateway** (e.g., 192.168.1.1 / 192.168.0.1), which represents the IP address of the local Wi-Fi router.

❖ **Accessing the Router Admin Panel:**

- Opened a web browser (such as Google Chrome / Microsoft Edge).
- Entered the Gateway IP address directly into the browser address bar: http://192.168.1.1
- Pressed **Enter** to establish a connection with the router's web configuration utility.

❖ **Verifying the Login Page:**

- The browser successfully rendered the router's administrator authentication screen requesting the login credentials (username and password).

2. Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

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IP Address, Subnet Mask and Default Gateway

I connected my HP laptop to my Wi-Fi network and used the ipconfig command in Command Prompt to check the network configuration.

Network Details:

- Device: HP Laptop
- IPv4 Address: 192.168.1.105
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

3.Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access. Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.

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I changed my HP laptop's IP address from Obtain an IP address automatically (DHCP) to a static IP address in the same network range as my Wi-Fi router.

Configuration:

- IP Address: 192.168.1.50
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1
- Preferred DNS Server: 8.8.8.8

After applying the settings, I tested the internet connection by opening a web browser and accessing a website. The internet connection worked successfully.

After completing the test, I changed the network settings back to Obtain an IP address automatically and Obtain DNS server address automatically to prevent future network problems.

4. In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

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1. Location of DHCP Settings:

- **Navigation Path:** Logged into the router admin interface and navigated to **Network Settings** > **LAN** > **DHCP Server**.

2. Current DHCP IP Range:

- **Start IP Address:** 192.168.1.100
- **End IP Address:** 192.168.1.200
- *(Note: This range allows up to 101 connected devices to automatically receive private IP addresses dynamically.)*

3. Impact of Duplicate IP Address Assignment (IP Address Conflict): If two devices are assigned the exact same IP address within the DHCP range, an **IP address conflict** occurs. As a result, both devices may experience intermittent or complete network connectivity loss, packet misrouting, and dropped connections because the router cannot determine which device should receive the incoming data packets.